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(i) **Fibrous joints or Synarthroses:** They are immovable fixed joints in which no movement between the bones is possible. Sutures of the flat skull bones are fibrous joints.

(ii) **Cartilaginous joints or Amphiarthroses:** They are slightly movable joints in which the joint surfaces are separated by a cartilage and slight movement is only possible. E.g., Joints of adjacent vertebrae of the vertebral column.

(iii) **Synovial joints or Diarthroses joints:** They are freely movable joints, the articulating bones are seperated by a cavity which is filled with synovial fluid. E.g., Pivot joint – between atlas and axis
Plane/gliding joint – between the carpals
Saddle joint – between the carpal and metacarpal
Ball and socket joint – between humerus and pectoral girdle
Hinge joint – knee joint
Condyloid or Angular or Ellipsoid – joint between radius and carpal

CTS-(Carpal Tunnel Syndrome) – The narrow passage (tunnel) bounded by bones and ligaments in the wrist gets narrowed and pinches the median nerve. This syndrome is mostly seen among the clerks, software professionals and pregnant women and people who constantly play or text in mobile phones.

An exhausted student was attending a lecture. After 30 minutes or so, he lost interest and he let go with a tremendous yawn. To his great distress he couldn't close his mouth –his lower jaw was locked open. What do you think would have caused it?

9.12 Disorders of muscular and skeletal system

(a) Disorders of muscular system

Myasthenia gravis: An autoimmune disorder affecting the action of acetylcholine at neuromuscular junction leading to fatigue, weakening and paralysis of skeletal muscles. Acetylcholine receptors on the sarcolemma are blocked by antibodies leading to weakness of muscles. When the disease progresses, it can make chewing, swallowing, talking and even breathing difficult.

Tetany: Rapid muscle spasms occur in the muscles due to deficiency of parathyroid hormone resulting in reduced calcium levels in the body.

Muscle fatigue: Muscle fatigue is the inability of a muscle to contract after repeated muscle contractions. This is due to lack of ATP and accumulation of lactic acid by anaerobic breakdown of glucose

Atrophy: A decline or cessation of muscular activity results in the condition called atrophy which results in the reduction in the size of the muscle and makes the muscle to become weak, which occurs with lack of usage as in chronic bedridden patients.

Muscle pull: Muscle pull is actually a muscle tear. A traumatic pulling of the fibres produces a tear known as sprain. This can occur due to sudden stretching of muscle beyond the point of elasticity. Back pain is a common problem caused